

How different is programming for hardware?

When programming the hardware, you're dealing with data obtained from the pins of the microcontroller. This is why it's necessary to ensure that you're using the correct pins to transmit or receive data. These pins must be configured for the task you want them to perform, before using them for transmitting/receiving data. All these processes can be summarised using the flowchart alongside.



Code for hardware devices is written in a modified form of C programming language called 'Embedded C'.

Connecting Arduino™ to your computer

After plugging the Arduino™ board into the USB port of your computer, you need to verify in the Arduino™ software that the:

1. correct board has been selected
2. correct port has been selected

Selecting the correct board

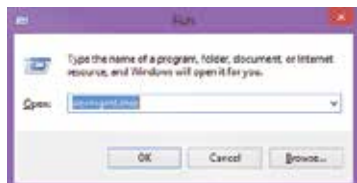
To select the correct Arduino™ board, from the menu bar of the Arduino™ software, navigate to Tools > Board and click on the board you're currently in the process of programming.

Selecting the correct port

A plugged in Arduino™ board is available as a device on the virtual communication port of the computer. The ID of the virtual port can be obtained from the Device Manager.

Use the following steps to open the Device Manager:

- ◆ Press **Win** (in Windows) and 'R' key simultaneously.
- ◆ In the text box that appears, type



Opening Device Manager